

Let's Get Personal

Defining and Estimating Effects of Dynamic Treatment Strategies Using Real-World Data

Wrap-up

Lucia Petito, Emma McGee, Rienna Russo

August 30, 2026

Why dynamic strategies?

- **Personalized medicine** is all about making *better clinical decisions* for individual people
- **Dynamic strategies** are a key component of personalized medicine because they help us answer *real-world questions*
 - “If HbA1c exceeds threshold X, then start medication A”
 - “If side effects occur, then switch to medication B”
 - etc.

How do we answer questions about dynamic strategies?

- **Randomized trials** are not always feasible, ethical, or available at the time we need for the population we care about
 - The number of clinically interesting dynamic strategies far outnumbers the trials that can actually be conducted
- **Real-world data** are therefore often needed

The utility of the target trial framework

- **Target trial emulation**
 - helps us clearly define the dynamic strategies and causal estimands of interest
 - E.g., the per-protocol effect of initiating and continuing (*unless changes a clinically warranted*) an SGLT2i vs. a GLP1-RA?
 - and avoids design-related biases (e.g., immortal time)
 - which are more prevalent when our interest is in dynamic strategies that depend on evolving individual characteristics

G-methods for estimating effects

- In practice, estimating the per-protocol effects of dynamic treatment strategies almost always requires **g-methods**
 - IP weighting of an MSM
 - G-formula
 - Doubly robust estimators
 - etc.
- Obtaining valid estimates requires not just appropriate methods, but also key **assumptions** and **high-quality data**

Summary

- Most real-world questions in pharmacoepidemiology are about **dynamic strategies** that are tailored to personal characteristics
- You now have a **conceptual** and **practical toolkit** to define and estimate the effects of these strategies using real-world data

Questions?

INTERNATIONAL SOCIETY FOR PHARMACOEPIDEMOLOGY
PRE-CONFERENCE COURSE 2026

Thank you!